WEEK 6 · DAY 2 · INSTRUCTOR ONLY

Quiz 4 Answer Key

Forecasting and remaining life · shuffled version

Instructor only. Built from the original `Time-series/MCQ/week-06-mcq.csv` (unchanged). Options were reshuffled with a fixed seed (6203) so the correct letters are spread out: multiple-choice answers sit at A ×6, B ×7, C ×6, D ×7 (in the original, most answers were option B). The 4 True/False questions keep their original order.

#	Answer	Correct option	What it tests	Level
1	**C**	The last observed value	Naive forecast	beginner
2	**B**	The same week one year (52 weeks) ago	Seasonal naive	beginner
3	**B**	On a typical week it was off by about 11 tickets	MAE in words	beginner
4	**C**	It repeated last year, when the network was much smaller	Seasonal naive came last	intermediate
5	**A**	Makes the forecast follow the latest week closely	Alpha in exponential smoothing	intermediate
6	**B**	False	Keeping a baseline	beginner
7	**A**	It must be known at the moment the forecast is made	The leak rule	beginner
8	**D**	df["tickets_raised"].shift(1)	shift direction	beginner
9	**D**	The 52-week lag does not exist for the first year	Rows lost to lag 52	intermediate
10	**A**	So week 52 sits next to week 1 as a smooth cycle, using two columns instead of 52	sin and cos	intermediate
11	**D**	6.75	Like with like	advanced
12	**A**	True	Coefficient sanity	intermediate
13	**B**	A range the true value should fall inside a stated share of the time	Prediction interval	beginner
14	**C**	forecast plus and minus 1.645 times the residual standard deviation	Building the band	intermediate
15	**C**	The interval is too narrow for a 90% claim	Coverage	beginner
16	**D**	The model was scored on weeks it had already been fitted to	Why too narrow	intermediate

#	Answer	Correct option	What it tests	Level
17	**B**	False	Turning up z	advanced
18	**B**	Most likely around 71, plausibly 63 to 79; expect bigger misses at season changes	Communicating	intermediate
19	**D**	The time until the trend crosses its operational limit	Framing RUL	beginner
20	**C**	The slope and intercept of the best straight line	np.polyfit	beginner
21	**B**	1 May 2025	Battery call	intermediate
22	**B**	A planning figure with a range, re-estimated every year	Long reach	intermediate
23	**A**	It falls in discrete steps at each station-keeping burn	Propellant shape	intermediate
24	**A**	True	Time to intervention	beginner
25	**A**	Forecasting from past dates using only what was known then, and scoring against what came next	What backtesting is	beginner
26	**C**	The model is refit on everything before the cut and forecasts the next few weeks	Walk-forward	beginner
27	**D**	The spread across folds and when the bad folds occur	Report the spread	intermediate
28	**B**	Naive won five of six folds with lower mean error	Baseline backtest	advanced
29	**A**	13 days early	Battery check	beginner
30	**D**	The warning time left to act shrinks	Accuracy versus lead time	intermediate

Marking it together (10 minutes)

1. Read out the letters; everyone marks their own paper.
2. Ask for hands up on each question. Any question that more than a third of the room got wrong, explain it again with the matching day's slide.
3. Map each topic back to its day: forecasting baselines → W5D2 · regression-based forecasting → W5D3 · uncertainty and prediction intervals → W5D3 · trend-to-threshold and remaining useful life → W6D2 · backtesting → W6D1 (Q29–30 → W6D2).

Quick topic tally

Topic	Questions
forecasting baselines	1, 2, 3, 4, 5, 6
regression-based forecasting	7, 8, 9, 10, 11, 12

Topic	Questions
uncertainty and prediction intervals	13, 14, 15, 16, 17, 18
trend-to-threshold and remaining useful life	19, 20, 21, 22, 23, 24
backtesting	25, 26, 27, 28, 29, 30